SpendWise: Expense Tracker

Application Name: SpendWise | Group Number: 2

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1. App Overview

SpendWise is a mobile expense tracker that makes it quick and simple to record daily spending and see where the money goes. Users add an expense with a title, amount, category and date in a few taps. The app summarises spending by category and month, and shows recent expenses and totals.

Problem addressed: Many people, especially students and young professionals, do not track spending and lose track of where their money goes. SpendWise offers a lightweight way to log expenses and understand spending habits.

Intended users: University students and young adults on a tight budget who want a simple personal finance tool.

2. Core Features

- Dashboard: total spending, recent expenses with category icons, and an insight card.
- Add, edit and delete expenses with title, amount, category and date.
- Predefined categories: Food, Transport, Shopping, Bills, Entertainment, Health and Other.
- Spending summary: totals by category and month, displayed with a basic chart.
- Local persistent storage, input validation and a clean Material 3 interface.

Optional features (only if time allows): Monthly budgets and warnings; receipt scanning with OCR; CSV export and dark mode; cloud sync and accounts (e.g., Firebase).

3. Data Requirements

Each expense stores a unique ID, title, amount, category, date and optional note. Settings include currency, selected categories and optional budget limits.

Database: Local on-device storage (SQLite via sqflite, or Hive). No account or server is required for the core version.

External API: Not required for core features; the app works fully offline.

4. Potential Challenges

- Data persistence: consistent records when editing and deleting expenses.
- State management: separating UI, database and business logic (Provider or Riverpod).
- Charts and summaries: grouping expenses accurately by category and month.
- Input validation: preventing invalid amounts, empty fields and incorrect dates.
- Team collaboration: avoiding merge conflicts with feature branches and pull requests.
- Scope control: starting optional features only after core features are stable and tested.

5. Planned Technologies

Flutter / Dart, Material 3, SQLite (sqflite) or Hive, JSON (dart:convert), Provider or Riverpod, a chart package (e.g., fl_chart), Git and GitHub.

6. Team Responsibilities (Step 1)

Coordinator: Khalil Khalilli — organizes meetings, distributes tasks and communicates with the assigned TA.

Member	Assigned task
Khalil Khalilli	Coordinate team, collect TA feedback, submit final PDF via SUCourse.
Harun Can Yurdagül	Write app overview, problem and intended users.
Akif Şentürk	Define core/optional features and review project scope.
Atilla Mammadzada	Write data requirements, challenges and technologies.
Berat Can Satır	Set up public GitHub repo, collaborators, README, .gitignore and initial commit.

7. GitHub Repository

https://github.com/cansatir/cs310-project